AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT132 – M132
Linear Algebra

Final Examination
Summer Semester 2025/2026

Time Allowed: 2 Hours

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Instructions:

- Total Marks: 50
- This exam consists of 3 parts.
- ALL questions must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Q–1: [5×3 marks] Choose the correct answer:

1.1: Which of the following is a basis for $\mathbb{R}^3$.

a) $\{(1,1,1),(1,1,0),(1,0,0)\}$
b) $\{(1,0,0),(0,2,0),(0,3,0)\}$
c) $\{(1,1,2),(1,2,1),(0,-1,1)\}$
d) All are bases
e) None of the above

1.2: Which of the following is true?

a) If S is a set of vectors in $\mathbb{R}^n$contains more than n vectors, then it is linearly independent
b) A set of n vectors in $\mathbb{R}^n$ is always a basis
c) If a square matrix has linearly independent columns, then it is invertible
d) Every subset of a linearly independent set is linearly dependent
e) None of the above

1.3: Let $T: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ is a linear transformation define whose matrix is$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$.

Find $T\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) =$

a) $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$
b) $\begin{bmatrix} 5 \\ 5 \end{bmatrix}$
c) $\begin{bmatrix} 4 \\ -5 \end{bmatrix}$
d) $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$
e) None of the above

1.4: If the matrix $\begin{bmatrix} 2 & 1 \\ 1 & a \end{bmatrix}$ has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 3$, then $a =$

a) -2
b) 2
c) 1
d) -1
e) None of the above

1.5: Consider the linear system $\begin{cases} 2x + y = 1 \\ x - 2y = 4 \end{cases}$. Which of the following is TRUE?

a) The system has a unique solution
b) The system has infinitely many solutions
c) The system is inconsistent
d) The system is homogeneous
e) None of the above

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Q–2: [8 marks] Answer the following questions as True or False (1 mark), Justifying your answer (3 marks):

2.1: If $A = \begin{bmatrix} -1 \\ 3 \\ -3 \\ 1 \end{bmatrix}$ with $(rA)^T(rA) = 20$, then $r = \pm 1$.

2.2: Every symmetric matrix is invertible.

Each question is worth 9 marks. Answer the following questions:

Q–3: [5+3+1 marks] Let $W = \left\{ \begin{bmatrix} a \\ a-b \\ c \end{bmatrix} : c = a + 3b, a, b \in \mathbb{R} \right\}$ be a subset of $\mathbb{R}^3$.

a) Show that W is a subspace of $\mathbb{R}^3$.
b) Find a set S such that $W = \text{span}\, S$.
c) Find a basis for W.

Q–4: [5+3+1 marks] Define the mapping $T: \mathbb{R}^3 \to \mathbb{R}^2$ by $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x+y \\ y-z \end{bmatrix}$.

a) Show that T is a linear transformation.
b) Find the null space $N(T)$, and the range $R(T)$.
c) Find $Dim(N(T))$ and $Dim(R(T))$.

Q–5: [5+4 marks] Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$.

a) Show that A has eigenvalues $\lambda = -1, 1, 3$.
b) Find an eigenvector corresponding to $\lambda_1 = 1$ and $\lambda_2 = -1$.

End of questions

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